

Agile coaching for global software development

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SUMMARY

This paper presents a case study on building a successful agile coaching team focusing on global software development projects in an international Nordic-based software company. We describe how the team of eight coaches was built, and how the coaches work together as a team using agile practices, such as weekly iterations, backlogs, and daily stand-ups. Furthermore, we describe their ‘deep and narrow’ approach to working closely with case projects, and the benefits to the coached projects. We describe the main challenges in building the coaching activities in a globally distributed company. The paper concludes with a set of lessons learned for coaching globally distributed projects and a discussion of the future plans for scaling up coaching by building a global coaching network within the company. We collected the data using 13 semi-structured interviews of the coaching team members, and personnel from four coached case projects. Copyright © 2012 John Wiley & Sons, Ltd.

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1. INTRODUCTION

The benefits and importance of coaching has been recognized in effectively adopting agile methodologies [1, 2]. The *agile coaches* working with the team can either be external consultants, or internal agile experts. Companies often use a combination of both, with external consultants doing initial trainings and helping to kick-start agile development, and helping to initiate the development of internal coaches who later take over the coaching [1, 3–5].

Although case studies and experience reports on agile coaching in general exists, the literature on how to coach globally distributed software development projects to adopt agile practices is scarce. In addition to the challenges encountered when adopting agile methodologies in collocated environments, global software engineering (GSE) poses additional challenges. Because of geographical distance, issues, such as the lack of face-to-face communication, time differences, and difficulties in building and maintaining trust increase the need for coaching, because the adoption of agile methodologies in such environments is far from straightforward.

In this paper, we describe how a large, global Nordic-based IT organization has built an internal coaching team for coaching globally distributed software development projects to adopt agile methods. More specifically, we describe the way the coaching team has been built, and how its members work together as a team. We describe the ‘deep and narrow’ approach to coaching taken by the team, and the benefits this approach provides to the coached projects. We discuss the challenges encountered in the organization while building the coaching team. Finally, we present a set of lessons learned for coaching

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globally distributed projects, and discuss the future plans for building a global coaching network to better support the future coaching needs of this globally distributed company while it carries out a company-wide lean and agile transformation.

The paper is structured as follows: Section 2 presents a brief literature review on agile coaching, Section 3 describes the research methods, Section 4 presents the results, and finally, Section 5 discusses the results, the limitations of the study, and proposals for future research.

2. PREVIOUS WORK

Adoption and sustaining of agile methodologies can be greatly facilitated by the use of agile coaching. An agile coach is an experienced user and teacher of agile methodologies, who can take on many roles, such as teacher, facilitator, coach-mentor, conflict navigator, collaboration conductor, problem solver, etc., to help teams adopt and improve their use of agile methodologies [2].

Although much of the focus in the literature has been on the mentoring and teaching aspects of agile coaches, there is also an emerging understanding of the 'life-coaching' role an agile coach sometimes needs to take [2], and attempts to develop coaching frameworks based upon sports coaching analogies [5]. The benefits of coaching include improved understanding of agile practices and ways of working [1], better teamwork [6], higher product quality, and lower overall project cost [7].

While an agile coach strives to teach a team to work independently as quickly as possible, at the organizational level, there can often be a need for permanent, or at least long-term, availability of coaches to help roll-out, support, and reinforce the use of agile practices. Thus, organizations often opt for developing internal coaches that combine an in-depth understanding of the organization, its customers, and processes with coaching skills and excellence in the use of agile methodologies. Internal coaches can be developed through co-coaching with experienced, sometimes external, expert coaches, and by providing coaching training [1, 3, 8].

Existing literature on agile coaching describes the roles and ways of working of agile coaches, experiences in building agile coaches, and coaching communities within organizations. A number of tutorials and workshops on agile coaching have also been offered at various conferences [9–15]. Although these reports are interesting, there is very little literature on the particulars of coaching and building coach communities for globally distributed software development projects [4].

3. METHODOLOGY

The research presented in this paper is based on a qualitative single-case study approach [16]. We opted for a case study because we wanted to get an in-depth understanding on how a coaching team works. The studied coaching team was selected purposefully, that is, by choosing an information-rich case [17], because we encountered an extremely interesting case from a company participating in an existing research program.

The case study organization is a globally distributed software company that provides software products and services. This Nordic-based company has development centers both in Europe and in Asia. Typically, the development projects are distributed between two or more locations. The studied coaching team is located at a large delivery center in Eastern Europe.

The main data collection method was semi-structured theme interviews. We also received documents, such as the internal wiki pages of the coaching team. The researchers and case study company representatives discussed the interview goals and the selection of the interviewees. Because the goal of the study was both to understand how the coaching team works and the results of the coaching, we interviewed coaches, and personnel from coached projects. An overview of the studied projects is shown in Table I.

The final selection of the interviewees was made by the coaching team. We interviewed four coaches, including the coaching team leader, and eight persons from four coached projects, including two persons per project. The interviewees from the coached projects were persons who had worked closely with the coaching team members, for example, project managers, release project

Table I. Case project overview.

Case	Printing	Banking	Telecom	Energy
Customer industry	Newspaper	Banking	Telecom	Energy
Project type	New product development	Bespoke application update	Bespoke application development and maintenance	Legacy software product development and maintenance
Distribution (# personnel min-max)	Nordic (6–17), Eastern Europe (4–19)	Nordic (6–15), Eastern Europe (6–10)	Nordic (7–15), Eastern Europe (10–16)	Nordic (15–30), Eastern Europe (6–30)
Project length	2004–2009	2004–ongoing	2007–ongoing	1980s–ongoing (GSD since 2009)
Coaching period	2007–2009	2008–ongoing	2007–2010	2009–ongoing
Situation when coaching started	Challenged project	Challenged project	New collaboration starting with a Nordic customer	Challenged project, transfer to other site planned
Process change	From waterfall to iterative Rational Unified Process (RUP)	From waterfall to iterative Rational Unified Process (RUP)	First Scrum, then ‘Scrumban’	From waterfall to Scrum

managers, and senior developers. In addition, we interviewed the former manager of the coaching team, whose task had been to expand the coaching activities. The interviews lasted 1–2 h per person. Moreover, the researchers participated as observers in one of the coaching team's weekly Monday meetings and in one of their daily stand-up meetings. The coaches also arranged a session during which the coaching team presented their way of working to the researchers. Both researchers participated in all the interviews, one being the main interviewer, the other presenting supporting questions and taking notes.

The interviews were audio recorded and transcribed. The first round of analyses was based on the interview notes, from which categories were created. The transcribed interviews were subsequently coded using these categories, and quotations were extracted.

To validate the findings, we sent the final paper for checking to the interviewed coaching team members, and to their former manager, whom we had interviewed. We received a few minor suggestions for improvement that we incorporated into the text.

4. RESULTS

4.1. History: building the coaching team

At the time of the study, the coaching team had existed for 5 years, having grown from two to eight coaches, and had coached over 50 projects. Half of the projects had a long-term relationship of over a year with the coaches, while the other half consisted of project assessments followed by improvement suggestions.

The coaching team was initially built around two active individuals who had been working on a successful globally distributed software development project applying agile practices. This project had been so successful that many people outside the project had heard success stories and started asking advice of these two central persons, an off-site project manager, and an architect/development team leader, who had participated in tailoring the process for this project.

'So [the head of one major department] asked: "Could you please tell your story, your experiences, your practices to the other cases? So they could improve." That's how it started.'— *A Coach*

These two persons started giving training sessions, sharing their experiences on good global software development and agile practices, while still continuing to work for the successful project. However, the 20% time allocation they had for coaching turned out to be clearly insufficient, and the coaches left the project, forming a coaching team to which two other persons were recruited. Thus, a team of four full-day coaches was created in 2006. At that time, agile methods were not well known in the company, thus the team focused on arranging agile trainings.

'So we had those trainings, they consumed lots of time, and we had more and more requests coming [from projects]. We tried at least to give them advice, so we were consulting rather than coaching at the time. It was only a few cases [in which] we focused deeply, [using] this deep and narrow approach. And at some point we decided that this is not the way.'— *A Coach*

During 2007, the team grew from four to eight coaches. The development of these new hires from novices to competent coaches took from 6 months to 1 year depending on their previous experience. The new junior coaches were trained by working in pairs with experienced coaches.

The coaching team developed and started to practice a 'deep and narrow' approach to coaching, meaning that the coaches aimed to work closely with their case projects to solve targeted problems.

So far, the coaching team has not marketed their services, but they are constantly offered more work than they can take on. Most case projects approach the coaching team as a result of people participating in the training sessions, or someone having heard about the team and their results. Management helps the team to select which cases to work on.

The coaching team has also received company-wide recognition: they were awarded a company-level award for their achievements. Currently, a company-wide coaching network is being formed to help disseminate the best practices for coaching globally distributed projects.

4.2. Coaching process and practices

The coaches work together as a tightly connected team, which is something they find crucial to their success. Over time they have built many practices for working efficiently as a team. Next, we present these practices, which are illustrated in Figure 1.

4.2.1. Team leader. Although the coaching team views itself as a ‘team of leaders’, they have recognized the need for a team leader. The leader is the most senior coach, around whom the whole team has grown. Earlier, he had been the primary coach for many cases, but after the team grew, they noticed that there was a need to put more effort into their own team. Thus, the team leader has now dedicated half of his time to facilitating the work of the coaching team.

4.2.2. Working in pairs. The coaches aim to work in pairs whenever possible. Training sessions are normally held by two coaches. A pair of coaches is chosen for each case project: a primary and a secondary coach. The main reason for working in pairs is to be able to discuss the challenges of the case projects together. A pair for each case is chosen according to the competencies of the coaches, ensuring that their competencies are both a good fit with the task and complementary.

When the coaching team was growing fast, pairs were formed of a senior and a junior coach to teach the juniors. However, this was deemed insufficient to ensure the fast growth of the juniors. Thus, the juniors were given important responsibilities early on, for example, to lead sessions, and the senior coaches would give them feedback afterwards. Moreover, the team leader started monthly ‘goal development discussions’ with the juniors. After a year, the juniors had reached adequate competence levels, and the division between seniors and juniors was no longer needed.

Changing pairs for different assignments ensures that knowledge spreads within the team. Each coach is normally a primary coach for one and a secondary coach for a couple of projects. However, this depends on the coaching phase and on the amount of coaching needed in each case.

4.2.3. Weekly iterations. Because the coaching team teaches agile and lean ways of working, they consider it important to use the same practices inside their own team. The coaching team uses weekly iterations, starting on Mondays with an ‘Iteration Planning and Assessment Meeting’, and ending on Friday afternoons with a ‘Retrospective Meeting’.

The coaches have coaching backlogs for each case project. When making decisions, they vote because they feel voting is more efficient than ending up in possibly long discussions. The team is physically located in a team room, which they use when not visiting customer projects.

During the week, the coaches are busy working face-to-face with members of the case projects. Especially during the mentoring phase, the coaches find it important to physically spend as much time as possible with the cases. However, they have reserved early mornings from Tuesday to Thursday for coaching team stand-up meetings.

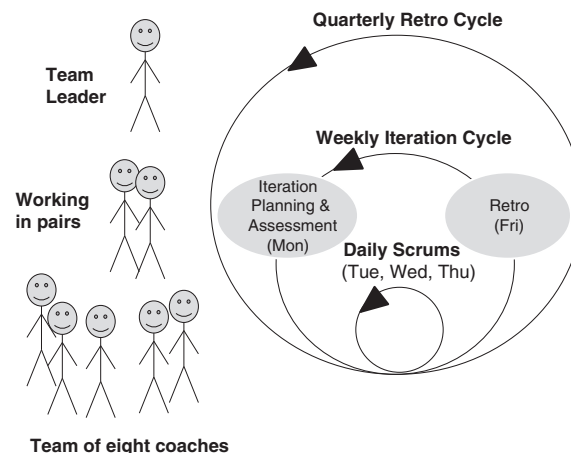


Figure 1. Working practices of the coaching team.

The team plans and records their working hours. Each coach spends approximately 20 h per week on case projects. A primary coach uses 10–20 h for his primary case and each secondary case receives a couple of hours. The rest of the weekly working time is spent on trainings, learning, and team activities. The coaches monitor their workload, and draw burn-down charts.

4.2.4. Daily meetings. The coaching team has stand-up meetings at 8:30 AM, except on Mondays and Fridays when they have other common team meetings. The purpose of the daily meetings is to close completed work items, discuss new unexpected events and new issues of the day, and collect items to wiki for the Iteration Planning and Assessment Meeting and for the Retrospective Meeting.

4.2.5. Iteration planning and assessment. On Monday mornings, the coaching team has its regular weekly Iteration Planning and Assessment Meetings, during which the coaches go through the objectives and work items, both completed and planned. While showing items from the backlog, each coach describes what has happened during the previous week and what will happen during the following week. The workload, burn-down chart, and emotions are reported, and external changes and risks are assessed.

4.2.6. Weekly retrospectives. Every Friday afternoon, the coaching team meets face-to-face to recap the past week. These sessions have two main topics: ‘to be discussed’ and ‘knowledge sharing’. Topics for each of these categories are collected to wiki during the week, when coaches face issues they would like to discuss within the team or when they have learned something new that might benefit everybody on the team. Without a dedicated time slot, this would be impossible during the busy week.

Every coach is expected to also share positive news in the retrospective, for example, positive customer feedback. Before this ‘positive’ practice was established, the meeting could end up being quite depressing, with everybody sharing their problems. The team decided to change this; just before the weekend, it is nice to bring up positive things!

4.2.7. Quarterly retrospective. The quarterly retrospective is also a team-building event, during which the team spends time together outside the office, for example, by heading for the mountains.

4.2.8. Common way of working with case projects. On the basis of their experiences, the coaching team has developed standard steps for working with case projects. Even though each project is different, these steps seem to suit most quite well. The steps are presented in more detail in Section 4.3.

Constantly collecting and sharing experiences from the case projects and discussing the challenging cases within the coaching team is one of the most important practices of the team. This ensures that the whole team learns from all the cases and that the best possible knowledge from the team can be combined when suggesting solutions to the case projects.

4.2.9. Benefits of working as a team. According to the literature, many coaches work quite independently, often as part of an agile project team. Some might be part of a loosely coupled virtual agile coaching network.

The interviewed coaches highly valued working as a coaching team. They mentioned several benefits of belonging to a team: a sense of belonging, getting help, having discussion partners, sharing experiences, being able to choose the best possible coach for each case, and being able to balance the workload between the coaches. Several coaches also pointed out that their individual learning and growth as a coach would not have been as dramatic without the supporting team.

4.3. Coaching approach

The coaching team uses a ‘deep and narrow’ approach, which means that they prefer taking long-term responsibility of each case project. They help the project to identify improvement needs and create an improvement roadmap. They support the project at each step of the roadmap, until the project is ready for independent continuous improvement.

In contrast to the deep and narrow approach is a ‘broad and shallow’ approach, which the coaches used earlier for some cases, when their resources did not allow for deep long-term commitment. The broad and shallow approach includes trainings, a workshop, and recommendations for the project to follow on its own. The broad and shallow approach is certainly better than nothing; however, the

coaches feel that the results with the deep and narrow approach are much better, thus they prefer it. In the deep and narrow approach the coaching team supports the case project for approximately a year. A few long-term cases have received their attention for as long as three years.

Next, we present the steps the coaching team uses when following the deep and narrow approach. The steps are illustrated in Figure 2.

4.3.1. *Receiving request.* Coaching starts when the team receives a request from a project.

At the time of the interviews, the coaching services were free for the projects because the coaching team was part of a corporate support function.

Most of the projects contacting the coaching team had heard success stories of previously coached projects, or the contacting person might have worked in a previously coached project.

‘The way people find us and contact us is mainly relationship-based. So, it’s typically a project manager who I coached on a previous case, and he asks me for help, stops me and says: “Hi, this is a new case, 40 people in [off-shore location] overnight and 15-year-old development in [onshore location], and it [migration of work] should happen in less than six months.” And I say, just a second, we need more time to discuss this case. And then it’s more than one year of coaching. . .’ — *A Coach*

Typically, the project contacting the coaching team is in serious trouble, for example, they cannot meet their deadlines and have huge quality problems. A typical project is geographically distributed among several countries, and is working with a fixed schedule and scope. Because the resources for coaching are limited, the higher managers of the coaching team make the final decisions regarding which cases to select, based on their business criticality.

4.3.2. *Mapping phase.* Once a case project is approved, the coaches start collaboration with the project in what they call a ‘mapping phase’. First, the coaches identify the key stakeholders of the project, establish a relationship with them, and ensure executive support. The coaching team has noticed that executive support is extremely important for the coaching to succeed. The projects in this company often start with a core team in a Nordic country and, only after a while, the project is distributed. The coaches learned through trial and error that they need to have executive support from this core site first. Earlier, they had reacted on support requests coming from their own offshore location and helped the local teams there. However, that easily led to problems when the suggested improvements were implemented only in the local teams and, at the same time, had to be ‘sold’ or at least approved by the core team in the other location, which caused resistance.

Second, the coaches will identify the needs of different stakeholders, and learn the daily rhythm of a case project and its way of working. Earlier, the coaches achieved this knowledge by conducting a number of interviews with the case project personnel after which they prepared a map of the current flow for the ‘Kaizen workshop’. Nowadays, they have left this interviewing activity partially or totally out, while discussing the same topics together with the project personnel in a Kaizen workshop that now takes two days instead of only a few hours as before.

Over time, the coaching team has developed a standard format for an improvement workshop, which they call a ‘Kaizen workshop’, using lean terminology, where ‘Kaizen’ means ‘continuous improvement’. In the current format, the workshop takes two days. The idea of the workshop is to invite persons with different roles around the same table, discuss the improvement needs from different viewpoints, and create and agree on a roadmap for improvement.

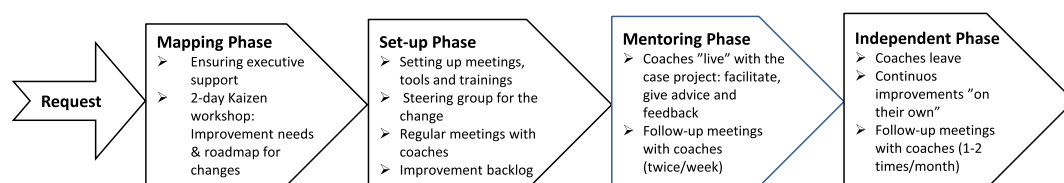


Figure 2. Coaching phases.

A critical aspect for workshop success is to involve all relevant stakeholders and representatives from all involved distributed sites, ensuring the participation of mixed roles bringing different perspectives, persons having decision power, and persons motivated to change things. It is important that there are business people who can tell others about the business goals, sales people who know how the project is sold to the customer, persons who know the system like architects and experienced developers, testing people, people from configuration management, and of course, a project manager or a release project manager and/or a project owner. The coaches select the participants together with their primary contact, that is, the person who ordered the workshop or the project manager/project owner:

‘We need people who are dissatisfied with the situation but still ready to accept an optimistic picture of our common future, willing to contribute and take ownership. We try to balance the best coverage of areas at the same time we keep the number of participants relatively small, usually 10–15. People from different sites usually travel to the same location to conduct face-to-face workshop. I wouldn’t go for a distributed workshop, since we usually touch sensitive topics we need to create a trustful and open atmosphere, and that’s almost impossible via video or phone.’— *A Coach*

During the first day of the workshop, the focus is on mapping the flow, and identifying the current problems and their root causes. In the morning, 2–3 h is received for mapping the flow, during which the coach asks questions leading to current problems. The problems are prioritized. After lunch, the root causes will be identified by asking ‘five whys’.

The second day of the workshop concentrates on solutions: identification of the improvement steps and designing the future flow. In the morning, 2–3 h are used for brainstorming solutions, for example in small groups. In the afternoon, the Kaizen steps are identified by brainstorming. The idea is that the case project teams can start implementing these small improvement steps immediately after the workshop. The steps and actions are agreed upon and documented. A responsible person is assigned for each solution.

The coaches have found these workshops useful for starting coaching activities because they unify the views between the different stakeholders and collaborating sites on the current flow and the biggest challenges and their root causes, and changes to be implemented. After the workshop, the project personnel has a detailed improvement roadmap and steps to start with. The coaches reported that, when seeing the bigger picture and the views of different stakeholders and sites, the workshop participants realize something that they have not understood before, as the following example shows:

‘The customer refused for several months to hire some testers, because they didn’t have money and time to educate them. After three hours at Kaizen workshop, they could clearly see how expensive this decision was, and how silly it is to stick to that. So they . . . the project owner left the meeting and called [the case company] to hire some testers immediately after that first session.’— *A Coach*

After the workshop, the results are compiled and the project management gives a go/no-go decision for the coach support when following the road-map created in the Kaizen workshop.

4.3.3. Set-up phase. In the set-up phase, the coaches take care that all necessary meetings, tools, and trainings are set-up to support the following of the roadmap created in the Kaizen workshop.

Regular meetings, for example twice a week, between the coaches and the key project personnel, such as managers and ‘change agents’, are scheduled. The change agents are individuals, for example managers or senior developers, who really believe in the change and are willing to lead the change from inside the case project.

The coaching/improvement project needs a steering group that meets regularly and follows the progress, makes high level decisions, and provides executive support.

The improvement project also needs a backlog for the tasks. This backlog can be set-up as part of the development project backlog.

The project personnel most probably needs trainings on agile practices to unify their knowledge and terminology, and trainings for specific roles, before starting to implement the changes.

4.3.4. Mentoring phase. The mentoring phase includes daily support from the coaches: facilitation of activities, such as planning, estimation, retrospectives and daily meetings, and support for problem

solving and answering questions. The coaches 'live' with the case project: they participate in the meetings, giving advice and feedback afterwards. The regular follow-up meetings, which occur twice a week between the coaches and the key personnel, offer a forum for monitoring, getting early warnings, and risk management, and opportunities to ask questions and receive feedback.

4.3.5. Independent phase. When the project can continuously improve on its own and does not need constant support from the coaches, it is time to move to the independent phase. This phase includes follow-up meetings once or twice per month between the coaches and the key stakeholders. These 1-h meetings offer opportunities for the project personnel to ask questions and for the coaches to ensure that the project is on the right track.

4.4. Benefits of coaching.

The persons interviewed from the coached projects mentioned a large number of benefits that the projects had gained from coaching. Next, we will summarize those benefits.

In all of the interviewed projects, the coaches had suggested process improvements toward agile methods. The interviewed managers felt that they were normally too busy taking care of daily tasks to lead the change at the same time without the outside help that the coaches provided. Thus, the help from the coaches made it possible to implement the changes.

The interviewees stated that, without coaching, it might be possible to make process improvements, for example the change towards agile practices, by following documents. However, several interviewees felt that, without coaching, they would not know if they are implementing the change correctly or if there are problems somewhere.

'(. . .) at the very beginning, it was that I had somebody I could ask whether I was doing things wrong or right.'— *A Coached Team Member*

According to another interviewee, without coaching, the changes would take much longer, because you have to first understand the new things and then to struggle on your own:

'I would say maybe it takes four, five times longer. (. . .), because of the project size, because you have to learn how to do it, but even then you are not sure if this is the correct way, then there are misunderstandings, then you have to learn based on your failures, and it can be even more than five times, so. . . '— *A Coached Team Member*

Coaches can help in 'selling' the changes to the stakeholders, for example, to the management, the external customer, or the distributed sites, based on positive experiences gained from other projects.

When planning the changes, the coaches can bring an independent, external point of view. They can help in unifying the stakeholders' views on important improvements, and locate the most important improvement targets and suggest suitable solutions:

'...because they are able to come and quite quickly locate what are the biggest issues that should be solved as soon as possible. (. . .) Not only detecting problems, because quite often we have some idea that something is wrong, but you don't know what to do with this. So, they were coming and saying, "You can either use this or that, and we think that this would be better for you, because of this" and they can just give you some arguments.'— *A Coached Team Member*

The coaches can help in designing the steps to the targets, and support the teams in implementing the practices, for example, by being present in the first few daily meetings and giving feedback after the meetings. The coaches are also available to answers questions.

The coaches have experience from several projects, so they can better anticipate the consequences of certain actions than a project manager, who is participating in the change for the first time:

'I think that without those coaches, of course we would try to find a way how to fix the things ourselves, but if you don't have experience from tens of projects and you are not specialized on the processes and so on, it's hard to see all the consequences of the ideas you have.'— *A Coached Team Member*

The interviewees commented that a coaching team supporting the primary coach is much better than just a single coach working without team support, because the combined knowledge of the team can

provide valuable input to the coach. Another benefit of having a coaching team instead of single coaches mentioned by our interviewees was that different coaches have different specialization areas. Together, they have a diverse set of skills; thus, different coaches may help in different areas.

The coaching and the implemented improvements helped many projects to be more successful than they would have otherwise been:

‘(. . .) we were in red numbers (. . .) and then it switched, and at the end, the project was very successful on all, (. . .) the customer was very satisfied.’— *A Coached Team Member*

The coaching may have even saved some projects, like one case project representative described, when we asked what the project would look like if the coaches had not been involved and helped the project in implementing the agile methods:

‘So, [the project would be] having a real waterfall (. . .) and the quality would be really bad, and the customer would be so disappointed that there wouldn’t be a chance of starting phase four. I think we would have ended the project in April last year [one year ago].’— *A Coached Team Member*

Even when selling a new project to an external customer mentioning that the project has coaches who can help the project to set up an efficient process smoothes the way and creates trust between the company and the customer.

4.5. *Lessons learned*

In Table II, we summarize some of the lessons that the coaching team has learned over the years on how to coach globally distributed projects.

4.6. *Challenges*

In this section, we describe the challenges and questions that the studied coaching team has faced. For some of the questions, the interviewed coaches had answers or at least opinions, but most of the questions were at least partly open.

4.6.1. *Deep and narrow or broad and shallow?* Because coaching resources are limited, there is always a choice that has to be made: How much coaching is enough for a project? For some projects a little bit of help could make a huge difference, for example, a training, whereas other projects might require day-to-day coaching for a longer period of time.

If the broad and shallow approach is chosen, the coaches could provide help for many more case projects than when working with the deep and narrow approach. The question is how to optimize the use of the limited coaching resources: Is it better to provide a little help to a large number of projects, or in-depth consulting to a few seriously troubled ones? The interviewed coaches answered that, according to their experience, the deep and narrow approach would bring more benefits than the broad and shallow approach. The team also considered the possibility of using both and choosing the approach depending on the situation in the case project.

4.6.2. *Choosing the cases.* Another choice involves which projects to coach of the ones requesting help. Should there be some company-wide policy for choosing the coaching case projects, for example, based on optimizing the potential benefit of coaching for the company or minimizing the potential losses in the projects? Who should make this choice? Some named managers or the coaching team?

4.6.3. *Business model.* At the time of the study, the coaching activities were free of charge for the coached projects. The organization had hundreds of projects, all accountable for their results. An open question was whether, in the future, projects should pay for coaching or not.

In the interviews, we asked the representatives of the coached projects whether they would have hired the coaches if they had to pay for them. The answers varied. For example, one interviewee answered that he would not have hired the coaches in the beginning, if he would have had to pay for it. Now, after seeing the results, he knows that he definitely should have hired them, even if there had been a cost attached.

Table II. Lessons learned.

Aim for a long-term relationship	• Long-term coaching relationship is beneficial. • Regular contact with the coaches provides quick solutions to problems and supports continuous improvement. • Challenge: Managers are often too focused on short-term benefits.
Ensure executive support from all sites	• Management support and commitment from all the sites is crucial. • Besides the costs of coaching, indirect costs—in particular, time for improvement work—must be allocated.
Involve everybody from the case project	• Project team members need to know what the change is, and why it is carried out. • Involve as many as possible in trainings and workshops.
Involve coaches in “selling” the changes to the other sites and to management	• Coaches, experienced from several projects, are good at explaining why changes are needed, what they mean in practice, and what the consequences are. • As outsiders, the coaches provide an objective view to the project.
Ensure the coaches have knowledge of successful GSE practices	• In addition to agile practices, supporting GSE practices are needed as well. • Agile practices need to be adapted to the distributed setting.
Involve coaches already in the planning phase of a new project	• Coaches can help in planning suitable GSE practices and in applying agile practices to a distributed project. • Unnecessary mistakes can be avoided.
Unify the views of different stakeholders by arranging a collaborative workshop	• A Kaizen workshop is a good tool for unifying stakeholders’ views: it opens the eyes of the participants to see the big picture of the whole distributed project and to recognize the root causes of problems. • Involve development personnel, management, and business people, and participants/representatives from all the distributed sites, in the Kaizen workshop. • Inviting everybody is preferable, but may not be feasible in big projects.
Find change agents in the project	• The change agent sees the need to make a change, believes in it, and is able to facilitate its implementation. • The change agent continues the work when the coaches leave the project. • A senior developer with a suitable mindset could be perfect for this role. • In a distributed project, all sites need their own change agents.
Agree on regular meetings with coaches	• Weekly/bi-weekly meetings with the coaches during the <i>mentoring phase</i> are beneficial to follow progress, discuss problems, and create solutions. • During the <i>independent phase</i>, monthly/bi-monthly meetings help ensure continuous change and rapid problem resolution.
The project team “owns” the change	• Ensure that the whole project team from all sites is involved and engaged in the change. • Make it clear from the beginning: the project team makes the changes and decisions, the coaches are there just to help and support them. • Even though the coaches support the project manager, they cannot do his or her job in managing the change or making decisions.

Coaching is a service, the financial benefits of which are difficult to measure and even more difficult to estimate beforehand, especially for a person who has no idea what to expect. Thus, having to pay for this service might cause the projects in most trouble to not take more financial risks by hiring coaches.

4.6.4. Identifying potential good coaches. When building a coaching team, new coaches need to be hired, either internally or externally.

The studied coaching team had developed their own requirements for coaches, and a way of recruiting. They considered it important that the whole team participates in interviewing a new candidate because, if chosen, he or she will be working with the whole team. The team had developed a list of skills that are important for a successful coach, among which leadership and communication skills were the most important, while a suitable technical background would help a

coach to understand the daily work of the case project personnel, and foster credibility among the technical personnel. Some of the coaching team members had started by working as part-time coaches, which they soon found did not work at all because both the project work and coaching easily required full-time commitments. All the interviewed coaching team members felt that, because they work very closely with the real projects and see many more projects as coaches than they would otherwise see, the fear of the 'ivory tower syndrome', that is, coaches becoming estranged from the real life of project personnel, is totally unnecessary. However, this fear had been raised by some of their managers, who themselves had already worked for a long time outside the real projects.

4.7. Future — towards a global coaching network

4.7.1. Why is a global coaching network needed? Most projects that the studied coaching team coached were globally distributed, with a majority of them distributed between two sites. However, the coaching team was located at one of the sites and, especially in the beginning, the coaching team coached mainly at their 'home site'. They soon noticed problems, as the biggest problems in the distributed projects arose from a lack of communication and collaboration between the sites and improving this situation could not be easily achieved by coaching only one of the sites. Moreover, implementing a new process model was not that useful or even possible if only one site was using the new process model (e.g., Scrum) while the other site was still sticking to the old process model (e.g., waterfall).

Thus, the coaching team soon understood that coaching all sites of a case project would be needed. However, that required quite a lot of traveling by the coaches and even this did not seem to be enough because, in the active mentoring phase, a coach needs to be present and help the teams face-to-face in their daily activities. Thus, a longer-term stay of a coach or coaches at the distributed site(s) would be needed, as was arranged, for example, in one of the interviewed projects. In the long run, long-term visits would be difficult to arrange because the coaches have families and cannot spend months away. Moreover, because this is a global company with hundreds of projects, one coaching team cannot coach all the projects and all the sites. The coaching team was constantly offered many more case projects to coach than they could take; thus, adding to the number of coaches was clearly needed. The coaches were hoping that, in the future, the different sites of each case project would have their own local coach or coaches. The local coaches from different sites could then collaborate closely.

4.7.2. How to grow the coaching network? According to the interviewed coaches, 'growing' a coach takes almost a year. This means that, when a suitable candidate is found, he or she should work as a junior coach with an experienced coach during that time. The coaching team members were planning to start locating and growing new coaches at the main distributed sites of this global company. These new coaches would work first as local coaches on the same case projects as the experienced coaching team members. The experienced coaches could visit the local junior coaches and help them with their local duties on the project and the junior coaches could visit the coaching team and work with them in pairs for a couple of weeks at a time. Later on, when there are a few local coaches at each site, these new coaches could then form their own local coaching teams, at least at the larger sites of this global company. All the coaches would then be part of a global coaching network that would share information and experiences across the sites. At the time of the interviews, the building of the global coaching network was just a plan, but since then the building has really started. At the time this article was written the interviewed coaching team, with help of their new coaching network members, are taking care of the company-wide lean and agile transformation that builds on the work the studied coaching team had successfully started.

5. CONCLUSIONS

5.1. Discussion

In this paper, we presented a case study on building a successful agile coaching team for coaching distributed software development projects in a globally distributed company.

5.1.1. Tight coaching team vs. loose coaching community. The paper showed how the coaches themselves worked as a team employing agile practices. Earlier coaching literature mainly presents loose coaching communities, while the studied coaching team was working as a highly integrated Scrum team. It seems that this approach worked very well for this small group of eight coaches. However, they saw that they had reached the limit of practical team size; further growing the team size would not fit to their way of working. Thus, for the future global coaching network, the coaches were planning to build tight local coaching teams, and a loose structure connecting the whole network.

5.1.2. Pure agile coaching vs. coaching troubled distributed projects. We described how the coaching team was gradually built to help challenged globally distributed projects to solve their problems by implementing both successful global software development practices and agile practices. In the beginning, the company's recommended process model was iterative Rational Unified Process (RUP); thus, the coaches helped the case projects to implement, for example, short iterations and frequent distributed meetings, and to improve collaboration between the sites by suggesting other suitable practices for global software development. Later on, Scrum, along with supporting global software development practices, became the standards the coaching team brought into troubled projects.

Compared with other coaching literature, the beginning of this coaching team was quite different: normally, internal or external agile coaches are asked to help when an organization or a project has already decided to implement some agile method and the organization might not otherwise be in any specific trouble. In our case study, the projects were most often in serious trouble; they were globally distributed, which was one major source of problems and, in most cases, the project itself did not have any specific plans for implementing an agile method. Rather, agile practices were most often part of the solution suggested by the coaching team.

5.1.3. Global software development coaches? We listed the most important lessons that the coaches had learned on coaching globally distributed case projects during the 5-year period the coaching team had existed. The successful practices for global software development described by the coaches are well in line with the current GSE literature. The main challenge is how to apply these practices in different projects. During the interviews, it became clear that the coaches had learned the contextual applicability of the practices through their work on previous case projects. Sharing this information between the projects and between the coaches made their learning faster, because the coaches had worked together only for 5 years and were already clearly very experienced in global software engineering. It seems that global software development projects could benefit from the help of 'global software development coaches'. Why is this kind of coach not used in other globally distributed software companies? We have not seen any reports on this in the literature, even though, based on this case, it looks like a beneficial practice. While studying global software development, we have seen that the same mistakes are normally repeated over and over again in new projects, even inside the same company. It seems that new project managers need to learn global software development practices by making their own mistakes first. Global software development coaches, who work in the same way as agile coaches by coaching distributed teams and their managers, might help in avoiding these mistakes and save companies huge sums of money.

5.1.4. Scrum is a good fit with distributed projects. When agile methods, especially Scrum, became more popular, the studied coaches started to help their case projects in implementing Scrum, while at the same time, coaching the projects on global software development practices. Coaches had learned that agile methods, especially Scrum, were a good fit with distributed software development because of the emphasis on and practices for frequent communication. One of the case projects utilized cross-site Scrum teams, while transferring the main part of the development of an old application to a new site. The distributed teams were efficient for knowledge transfer between the sites. However, most of the current literature suggests using site-specific teams for scaling Scrum in distributed projects with tasks that are not too tightly connected. This topic of using either distributed or site-specific Scrum teams seems to deserve further research to study the practices, benefits, and drawbacks, and the suitability of both types of teams, to different kinds of projects.

5.1.5. Deep and narrow vs. broad and shallow. We described the steps that the coaches used when they worked with their customer projects: mapping, set-up, mentoring, and independent phases. These steps seemed to work well. However, the length of the active coaching period seemed excessive, often lasting over a year, and involving several coaches. Thus, with this long-term intensive coaching model, the current coaching team is able to help only a fraction of projects contacting them.

In the future, building the planned coaching network would take time from the already busy coaches. Even though the coaches mentioned that the deep and narrow approach would bring better results, they saw that the future growth of the global coaching network would not be possible by coaching only a few projects at a time. The team considered having to move towards the broad and shallow approach with shorter coaching periods, for example, by finding internal coaches from the case projects who could coach the day-to-day activities in the projects, while the senior coaching team members could help them with bigger lines and by coaching the new coaches.

5.1.6. Global coaching network? We presented numerous benefits of coaching the case projects, and the main challenges in building coaching activities. As a last point, we discussed the future plans of this coaching team on building a global coaching network to help globally distributed projects and, as a new task, to tackle the challenges of making a company-wide transformation towards lean and agile practices. It will be interesting to see how this network will finally be built, how tightly it will work together and share knowledge, and whether it will be able to successfully transform the whole company, involving thousands of persons, to lean and agile. We have not yet seen this kind of a global coaching network in any other company or any descriptions of activities like this in the literature. Thus, the plans seem to be totally unique, and if successful, the lessons learned will be very valuable for many companies.

5.2. Limitations and future research

As a single case study of one coaching team working for one company, the generalizability of the results is limited. Although we interviewed half of the coaching team members and the rest participated in the common sessions, additional interviews might have provided additional insights. To capture the benefits of coaching on the projects, we interviewed representatives from only four coached projects, even though the coaching team had been working with over 50 projects. Moreover, these coached projects and the persons interviewed were chosen for this study mainly by the coaches and not selected independently or randomly, which might have introduced selection bias.

In the future, we hope to augment this study by also collecting quantitative data on the success of several coached and noncoached case projects from this same organization to be able to, at least on some level, quantify the benefits of coaching. Moreover, we hope to be able to follow how this organization is going to build the planned global coaching network to better support its globally distributed projects, and to perform the planned company-wide lean and agile transformation.

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